BKL one. BKL singularities are surely allowed by Einstein's relativistic laws. Garfinkle confirmed it by computer simulations. But more sophisticated simulations are needed to confirm that the BKL patterns of humongous stretch and squeeze do actually occur in the core of a black hole. I'm almost sure the result of those simulations will be "yes, they do occur." But I'm not completely certain.

A Black Hole's Infalling and Outflying Singularities



My physicist colleagues and I were pretty sure in the 1980s, as an educated guess, that there is just one singularity inside a black hole, and it's a BKL singularity. We were wrong.

In 1991 Eric Poisson and Werner Israel at the University of Alberta, Canada, working with the mathematics of Einstein's laws, discovered a second singularity. This one grows with time as the black hole ages. It's caused by extreme slowing of time inside the black hole.

If you fall into a spinning black hole such as Gargantua, lots of other stuff inevitably will fall in after you: gas, dust, light, gravitational waves, and so forth. This stuff may take millions or billions of years to enter the hole as seen by me, watching from outside. But as seen by you, now inside the hole, it may take only a few seconds or less, due to the extreme slowing of your time compared with mine. As a result, as seen by you this stuff all piles up in a thin sheet, falling inward toward you at the speed of light, or nearly the speed of light. This sheet generates intense tidal forces that distort space and will distort you, if the sheet hits you.

The tidal forces grow to become infinite. The result is an "infalling singularity" (Figure 26.11),[48] governed by the laws of quantum gravity. However, the tidal forces grow so swiftly (Poisson and Israel deduced)